npj | digital medicine

Article

Published in partnership with Seoul National University Bundang Hospital

https://doi.org/10.1038/s41746-026-02796-y

Reassessing negative 24 h pH impedance tests for hidden gastroesophageal reflux disease using multi feature anomaly detection

Check for updates

Songho Lee[1], Junhyeong Lee[1], Donggeun Park[1], Sang Kil Lee[2], Jae Hee Cho[3], Kyoung G. Lee[4] ✉, Hee Man Kim[5] ✉ & Seunghwa Ryu[1,6] ✉

Gastroesophageal reflux disease (GERD) diagnosis traditionally relies on acid exposure time (AET) obtained from 24-h multichannel intraluminal impedance-pH (MII-pH) monitoring, the gold standard for GERD diagnosis. However, a negative result (AET < 4%) does not always exclude GERD, as the limited 24-h monitoring window may fail to capture reflux events in patients with intermittent or low-frequency reflux. To address this limitation, we proposed a complementary machine learning-based framework targeting exclusively patients with negative MII-pH results (AET < 4%) to identify potential false-negative cases within this cohort, by integrating statistical and waveform-derived features from pH signals to enhance anomaly detection. Using one-class support vector machine and support vector data description models trained on real-world MII-pH datasets, the framework achieved an F_3 score of approximately 0.9 and identified potential anomalies undetected by the conventional AET criteria. Explainable AI techniques using Shapley additive explanations showed that features such as kurtosis and peak-to-peak amplitude contributed significantly to the identification of subtle reflux patterns within this cohort. These anomalies may indicate additional candidates for clinical reassessment within the AET-negative cohort. This complementary approach, operating downstream of the conventional MII-pH diagnostic system, could help identify potential false-negative cases among patients with negative MII-pH results, potentially assisting in their proper clinical management.

Gastroesophageal reflux disease (GERD) is a condition in which stomach contents reflux into the esophagus or mouth, primarily occurring when the pressure of the lower esophageal sphincter decreases[1]. This refluxate can damage the esophageal mucosa and lead to typical symptoms of GERD, such as heartburn and acid reflux. Furthermore, it may cause chest pain, dysphagia, globus sensation, and chronic cough[2]. Epidemiological studies indicate that GERD is highly prevalent worldwide, with reported rates of approximately 23% in North America, 17% in Europe, 5% in East Asia, 21% in the Middle East, 12% in Australia, and 23% in South America[3–5]. Over the last decade, there has been a significant increase in the number of patients with GERD, attributable to multiple factors such as the use of analgesics, consumption of certain types of food and beverages, smoking, high body mass index, and limited physical activity[6].

Accurate diagnosis of GERD is crucial given its high global prevalence and substantial clinical burden. The esophageal pH monitoring is a key diagnostic tool, providing an objective and quantitative assessment of esophageal acid exposure. Various tests, including catheter-based and wireless capsule-based pH monitoring, have been developed to measure gastric acid reflux, thereby improving the diagnostic accuracy. Recently, 24-h multichannel intraluminal impedance and pH (MII-pH) monitoring has been

[1]Department of Mechanical Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea. [2]Department of Internal Medicine, Yonsei Institute of Gastroenterology, Severance Hospital, Yonsei University College of Medicine, Seoul, Republic of Korea. [3]Department of Internal Medicine, Gangnam Severance Hospital, Yonsei University College of Medicine, 211, Eonju-ro Gangnam-gu, Seoul, Republic of Korea. [4]Center for NanoBio Development, National NanoFab Center, Daejeon, Republic of Korea. [5]Division of Gastroenterology, Department of Internal Medicine, Severance Hospital, and Yonsei Cancer Center, Yonsei University College of Medicine, Seoul, Republic of Korea. [6]KAIST InnoCORE PRISM-AI Center, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea. ✉e-mail: kglee@nnfc.re.kr; loverkorea2009@gmail.com; ryush@kaist.ac.kr

introduced to further enhance diagnostic accuracy and is considered the gold standard (GS)[7]. This device facilitates comprehensive detection of both acid and nonacid reflux by measuring intraluminal impedance and pH changes across multiple esophageal segments, thereby allowing more accurate evaluation of reflux symptoms. It also enables the collection of detailed physiological data, which can be used to calculate diagnostic indices and assist clinicians in making evidence-based decisions.

Alongside monitoring devices, some diagnostic indices have been proposed for the evaluation of GERD. One of the primary indices is acid exposure time (AET), which denotes the percentage of time during which the esophageal pH remains below 4. Based on AET values, GERD is classified as normal (<4%), inconclusive (4–6%), or conclusive (≥6%) according to Lyon consensus[8]. AET is highly reproducible and easily calculated, making it the most widely used parameter in MII-pH monitoring. Another major index is the DeMeester score (DMS), which is derived from 24-h esophageal pH monitoring and incorporates six parameters: A DMS greater than 14.72 is typically considered indicative of pathologic acid reflux. Calculating the DMS relies on the manual input of various patient activities, such as standing, lying down, and mealtimes. However, patients frequently omit these details, inevitably compromising diagnostic accuracy. Given these limitations, the Lyon Consensus currently prioritizes AET as the more reliable diagnostic metric, owing to its superior reproducibility and reduced dependence on patient-reported activity logs.

AET and DMS are well-established and clinically reliable metrics within the standard 24-h monitoring framework. However, because the monitoring window is inherently limited, patients with intermittent or low-frequency reflux may not exhibit sufficient acid exposure during the recorded period. As a result, a standard 24-h MII-pH study may yield false-negative (FN) findings, not because AET itself is inadequate, but because clinically relevant reflux events simply did not occur during the observation interval. In the present study, such missed yet potentially clinically meaningful cases were regarded as potential FNs[8]. To mitigate this limitation, clinicians may prolong the monitoring period beyond 24-h or incorporate auxiliary indices such as the symptom index, symptom association probability, mean nocturnal baseline impedance, and post-reflux swallow-induced peristaltic wave index. However, extending catheter-based monitoring exacerbates patient discomfort. Moreover, because these auxiliary indices require patients to manually record their symptoms and clinicians to match these records with reflux events, they remain partly dependent on subjective human input, which introduces variability and the risk of recording and interpretation errors[9,10].

To date, advanced approaches, such as using machine learning (ML), have been proposed to address these diagnostic challenges. ML-based anomaly detection (AD) algorithms can detect subtle patterns in high-dimensional data that traditional techniques may overlook. Previous research has highlighted the effectiveness of ML across diverse medical applications, ranging from imaging to physiological signal analysis[11]. For instance, Dairi et al. employed a variational autoencoder to extract features from blood test samples for COVID-19 detection and diagnosed anomalies using a one-class support vector machine (OCSVM)[12]. Hojjati and Armanfard proposed a deep autoencoding support vector data description (SVDD) model that integrates a neural network with standard SVDD, demonstrating its effectiveness across multiple domains, including medical data[13]. Beyond these ML-based AD models, autoencoder architectures based on 1D and 2D CNNs have also been applied to medical time-series and their time-frequency representations, where anomalies are detected as regions with elevated reconstruction errors relative to normal patterns[14]. More recently, Hu et al. proposed FECT-OSA, a transformer-enhanced multimodal supervised framework for contactless monitoring of obstructive sleep apnea that leverages event labels to distinguish apnea-hypopnea events from normal segments[15].

Despite these advances, ML and DL methods have rarely been applied to the re-evaluation of negative MII-pH results with the aim of identifying potential FN cases among patients with AET < 4%. Moreover, precise expert labeling of GERD MII-pH recordings is highly time-consuming and typically yields imbalanced datasets, in which abnormal patterns are both sparse and heterogeneous. Under these conditions, it is practically difficult to deploy supervised deep learning models that presuppose large, well-annotated, and balanced training datasets.

To address these constraints, we designed an unsupervised AD framework for reassessing AET-negative 24-h MII-pH recordings. Rather than relying on direct labels, the framework operates on AET together with statistical features extracted from 24-h MII-pH signals. The proposed AD framework combines the conventional clinical index AET with both statistical and waveform-derived features extracted from 24-h MII-pH signals. These features are used to train an ML-based AD model aimed at identifying potential FN cases. The framework specifically focuses on patients with AET values within the normal range (<4%) but near the diagnostic threshold, where subtle abnormalities may be overlooked. Finally, we assess the model's detection capability and the feasibility of subtle patterns indicative of potential FNs using clinical data, thereby providing clinicians with additional, interpretable information to support further clinical reassessment.

Results

Study design and overview

The overall reassessment workflow evaluated in this study is presented in Fig. 1. Patients presenting with reflux-related symptoms (Fig. 1A) undergo 24-h MII-pH monitoring using a catheter equipped with six impedance sensors and one pH sensor positioned in the esophagus. This setup records detailed physiological data, including multiple impedance channels and the pH signal (Fig. 1B). Conventional diagnosis basically relies on AET, which is the most robust diagnostic metric for GERD. However, subtle reflux episodes near the diagnostic threshold AET 4% may be overlooked, potentially leading to potential FNs (Fig. 1C).

To overcome this limitation, we introduce a multi-feature ML-based AD framework that integrates data-driven features with an unsupervised ML-based AD model. Initially, various data-driven features (e.g., mean, variance, etc.) were extracted from the raw pH signals. A correlation analysis was then conducted to identify a subset of informative features, including AET (Fig. 1D). These multidimensional features were used to train unsupervised ML-based AD models, specifically OCSVM and SVDD, with the intended use-case of identifying potential FN candidates within the AET-negative cohort (Fig. 1E). Finally, Shapley additive explanations (SHAP) were applied to quantify each feature's contribution to the anomaly decision, thereby supporting clinical decision-making (Fig. 1F). Accordingly, the target of the proposed framework is not the direct diagnosis of GERD itself, but the identification of potential FN cases among patients with AET < 4%. The task is therefore to detect subtle patterns indicative of potential FNs that may support further clinical reassessment beyond the conventional AET criterion.

According to the Lyon Consensus, esophageal AET < 4% is considered physiologically normal, 4–6% inconclusive, and >6% conclusive (abnormal)[10]. For Asian populations, Recommendation 13 of the Seoul Consensus classifies AET ≥ 4% as abnormal, thereby covering both the 4–6% and ≥6% ranges[1]. However, some patients continue to report reflux-related symptoms yet are not identified as abnormal because their AET values fall below this diagnostic threshold. Specifically, currently available sensing systems may not be sensitive enough to detect prolonged acid exposure during the standard 24-h monitoring period. Such cases may therefore denote potential FNs that are not detected by conventional AET-based criteria.

To show the representative cases, four examples of 24-h pH monitoring signals are presented, including a true positive (TP) case, a true negative (TN) case, and two potential FN cases classified as normal by AET (Fig. 2). Figure 2A shows the TP patient with frequent and sustained acid reflux (AET: 19.98%, DMS: 82.5), whereas Fig. 2B displays the TN patient showing stable signal patterns and accurately classified (AET: 0.00%, DMS: 0.8). Conversely, Fig. 2C, D present potential FN cases that were classified as normal based on AET, but show atypical patterns that may warrant

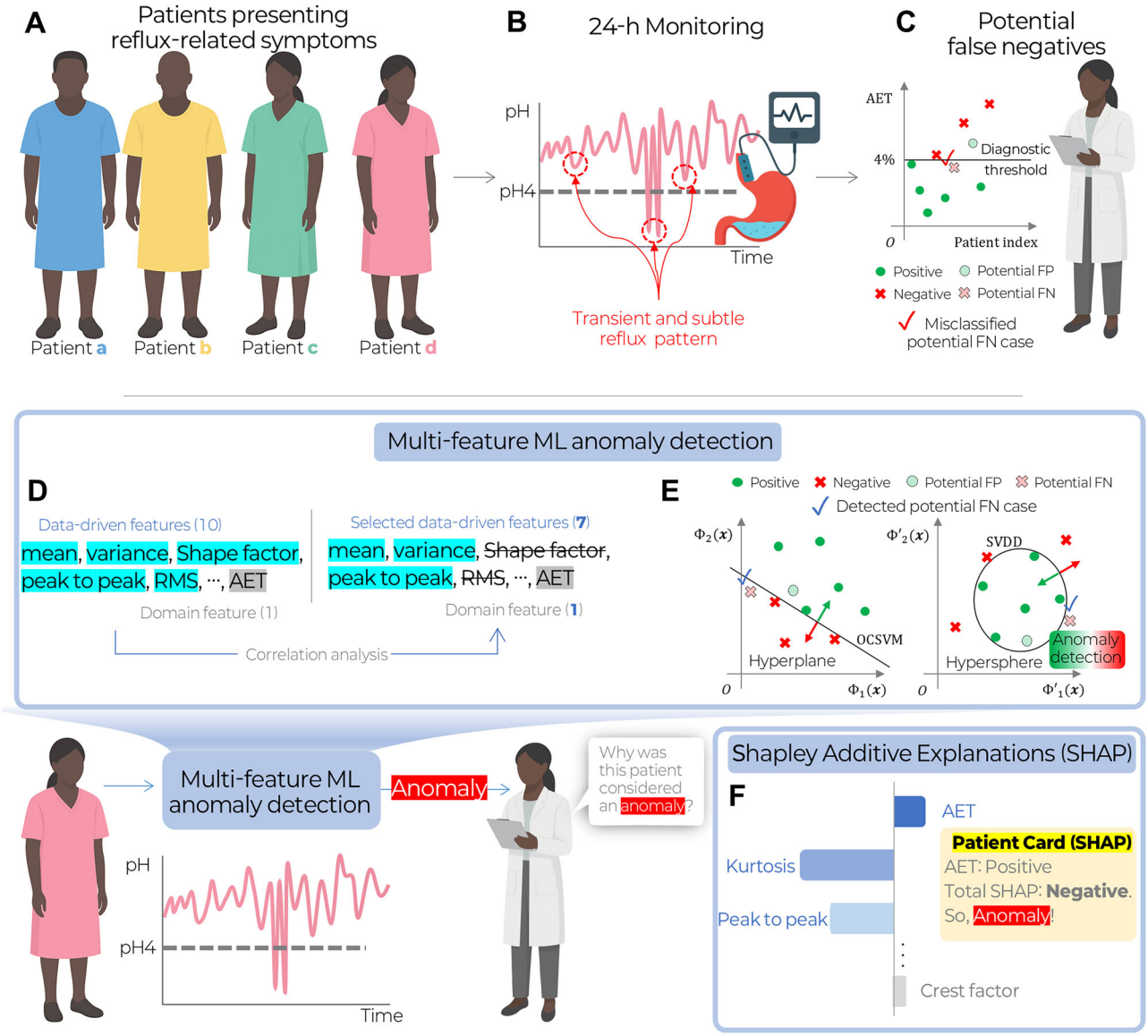


Fig. 1 | Overview of the proposed complementary framework for re-evaluating negative 24-h MII-pH results. The framework operates downstream of the conventional MII-pH diagnostic system, targeting exclusively the AET-negative cohort (AET < 4%) to identify potential false-negative cases that may have been missed due to the limited 24-h monitoring window. **A** Symptomatic patients (study cases including TP, TN, and potential FNs). The proposed framework focuses on the AET-negative subset. **B** 24-h MII-pH signals monitored from the esophagus. **C** Potential FNs within the AET-negative cohort whose reflux was not captured during the monitoring window. **D** Data-driven features extracted from raw MII-pH signals and the subset selected after correlation analysis. **E** Unsupervised ML-based AD in a multidimensional feature space. **F** SHAP-based explanation displaying each feature's contribution to the anomaly decision.

additional clinical reassessment. Figure 2C shows a low AET of 3.39% and a high DMS of 16.90 with SI < 50% and SAP < 95%, accompanied by a total of 103 reflux episodes (76 upright and 27 recumbent) and two recorded episodes of soreness, considered as a functional heartburn or potential FN case. Figure 2D shows a low AET of 0.58% and a low DMS of 2.40, with a total of 59 reflux episodes (37 upright and 22 recumbent). Nevertheless, symptom correlation analysis documented 695 belching events, including 284 episodes temporally associated with reflux (SI 40.9%, SAP 0.0%), indicating a considerable symptom burden despite normal AET/DMS values. In these representative recordings, sharp fluctuations and short-lived pH drops are observed without sustained low-pH exposure, which may contribute to their being insufficiently captured by AET alone.

Moreover, anomalies in clinical MII-pH data are inherently diverse, stemming from a range of sources beyond pathological acid reflux alone. These include sensor artifacts, signal distortions caused by inconsistent probe placement, and transient pH fluctuations influenced by dietary or behavioral factors. Given this heterogeneity and absence of detailed annotations distinguishing different types of anomalies, supervised learning approaches that require explicit labels are not feasible. Instead, our framework adopts an unsupervised approach by modeling the distribution of normal physiological patterns and identifying deviations as anomalies. To capture both true abnormal cases and potential FNs, we extract additional data-driven features from the pH signals, creating a multidimensional representation for AD using ML.

Feature extraction and selection

To enhance the detection of atypical reflux patterns that may be overlooked by conventional AET-based criteria, we augmented the feature set by extracting 11 features from the 24-h pH and impedance signals. The overall process is illustrated in Fig. 3, which outlines the workflow for feature extraction, correlation analysis, and final selection. As depicted in Fig. 3A, raw time-series data, including six-channel impedance (I1–I6) and pH

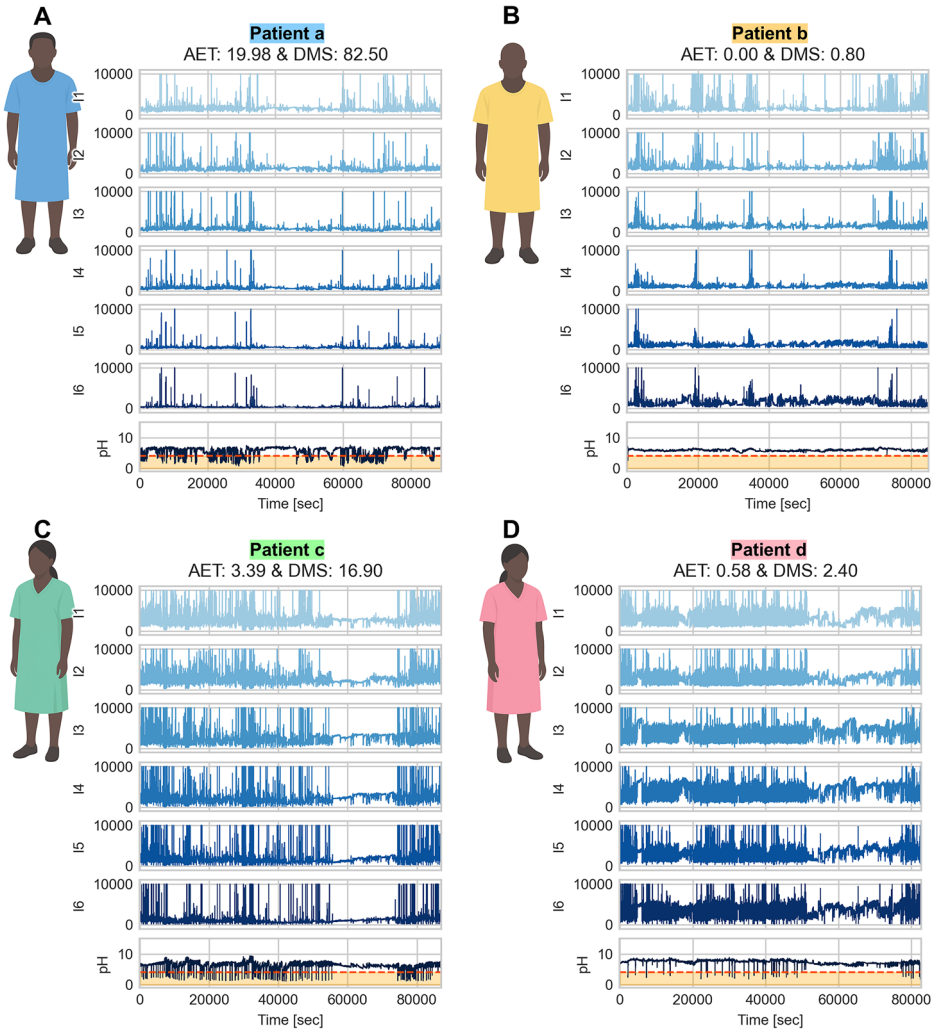


Fig. 2 | Representative 24-h esophageal MII-pH signals under AET-based diagnosis. A Case with AET: 19.98% and DMS: 82.50. **B** Case with AET: 0.00% and DMS: 0.80. **C** Case with AET: 3.39% and DMS: 16.9. **D** Case with AET: 0.58% and DMS: 2.40.

signals, were collected over a 24-h monitoring period. The dataset was sourced from patients at Severance Hospital, where MII-pH monitoring was performed under clinical settings (see "Data description" section for details).

From these signals, we derived 10 data-driven statistical and waveform features, including mean, variance, skewness, kurtosis, peak to peak, and root mean square (RMS), along with the domain-specific feature AET (Fig. 3B, see Table 3 for details). These features were designed to capture the amplitude and distributional characteristics of the signal and to represent recording-level waveform variations associated with transient, irregular, or weakly acidic reflux-like patterns that may not be sufficiently reflected by AET alone. Pearson correlation analysis was then performed to identify and eliminate highly correlated features, thereby mitigating the risk of overfitting[16]. The outcomes of the correlation tests are visualized in Fig. 3C. As the pH signal inherently does not have negative values, the absolute mean and mean were identical. Similarly, the RMS showed a significant correlation with the mean, as it is calculated as the square root of the average of the squared value of the signal. The shape factor, calculated as the ratio of RMS to the geometric mean, also showed a high correlation with these metrics. Therefore, the absolute mean, RMS, and shape factor were excluded, while the mean was retained for further analysis. Based on the above analysis, eight variables, including mean, variance, skewness, kurtosis, peak to peak, impulse factor, crest factor, and AET, were finalized as the major feature set for AD, as illustrated in Fig. 3D. The selected feature set could provide not only as a dimensionality reduction step, but as a clinically interpretable representation of acid waveform variability within the 24-h MII/pH signal. Representative signal patterns corresponding to the minimum and maximum values of each selected feature are provided in Supplementary Note 1.

Model training and hyperparameter tuning

To train unsupervised AD models, two different approaches, OCSVM and SVDD, were utilized, and their hyperparameters were optimized. These two models were selected for their ability to effectively detect abnormal patterns in high-dimensional feature spaces without relying on labeled datasets. This capability is particularly valuable in medical contexts, which are often characterized by imbalanced and incomplete labeling. Hyperparameter tuning is crucial for these models to balance detection accuracy with generalization and to prevent overfitting to the training dataset.

Figure 4 illustrates the conceptual framework and optimization outcomes of the selected AD models. Figure 4A contrasts the conventional AET-based classification with the OCSVM and SVDD models. In the AET method, each case is classified as either normal or anomalous based solely on whether the AET value exceeds the clinical threshold of 4%. However, this simple cutoff fails to capture subtle anomalies that are not fully reflected by acid exposure alone. In contrast, incorporating additionally augmented features $\boldsymbol{x}$ and high-dimensional mapping denoted as $\Phi(\cdot)$ or $\Phi'(\cdot)$, improves the ability to detect subtle patterns that may be overlooked by AET.

OCSVM identifies anomalies by constructing an optimal hyperplane that separates normal data from the origin in a high-dimensional mapped feature space, defined through kernel mappings such as $\Phi_1(\boldsymbol{x})$ and $\Phi_2(\boldsymbol{x})$.

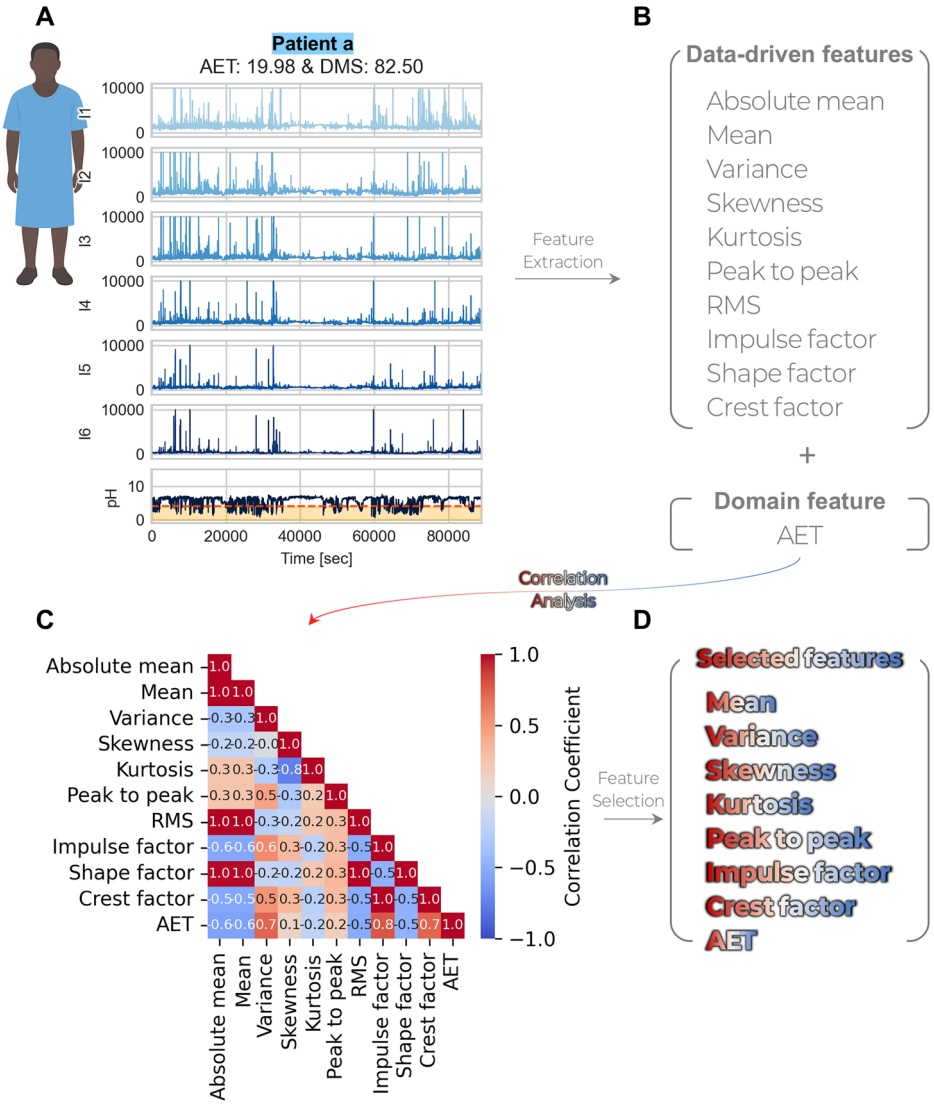


Fig. 3 | Workflow of feature extraction, correlation analysis, and feature selection. A Raw impedance (I1–I6) and pH data. **B** A total of 11 extracted features, including 10 data-driven features and AET. **C** Pearson correlation analysis of the extracted features, with color indicating correlation strength (red: +1.0, blue: −1.0). **D** Eight selected features used to construct the final preprocessed dataset for model training.

Similarly, SVDD encloses normal data points within the smallest possible hypersphere in the mapped space $\Phi_1'(\boldsymbol{x})$ and $\Phi_2'(\boldsymbol{x})$, identifying outliers as those located outside the hypersphere. Compared with the AET method, both OCSVM and SVDD offer flexible, nonlinear decision boundaries capable of detecting more nuanced and atypical reflux patterns. Their performance and generalization capacity critically rely on two hyperparameters: v, which controls the expected proportion of anomalies, and s, which defines the Gaussian kernel width and thus the complexity of the decision boundary.

To optimize model performance, a Bayesian optimization (BO) procedure was used to determine the best hyperparameters (v and s), as described in the Methods section. The full dataset ($n = 612$) was divided into a training set ($n = 489$, 80%) and a test set ($n = 123$, 20%). Hyperparameter optimization was then conducted using only the training set with 5-fold cross-validation to ensure robust generalization. The F_3 score was selected as the optimization metric, prioritizing recall over precision to better address the goal of minimizing FNs. The GS labels for F_3 score calculation were defined based on the AET criterion.

The BO results for the OCSVM model are detailed in Fig. 4B–E. Figure 4B illustrates the evolution of the F_3 score across 520 optimization iterations, beginning with an initial 20-iteration Latin hypercube sampling (LHS) phase, followed by 500 iterations guided by the expected improvement (EI) criterion. The optimal F_3 score achieved was approximately 0.912. Figure 4C visualizes the distribution of the evaluated hyperparameters ν and s, clearly distinguishing the initial random samples from the final optimal parameters. Figure 4D, E shows the convergence trajectories of the hyperparameters v and s, highlighting the optimal values ($v^* = 0.833$ and $s^* = 2.500$). The SVDD model underwent a similar optimization process. The detailed optimization results for SVDD, which showed patterns similar to those of OCSVM, are provided in Supplementary Note 2. Additional methodological details, including the formulations of OCSVM, SVDD, and BO, are provided in the "Methods" section.

Comprehensive evaluation of AD performance and feature contributions

First, the overall classification results for the entire cohort are summarized in Table 1 as a confusion matrix comparing the AET-based labels with the OCSVM predictions. The model identified 118 of 119 AET-positive cases as anomalies and additionally flagged 108 patients (21.9%) within the AET-negative group. Additionally, the relationship between the AD model classifications and conventional GERD phenotypes was examined. Among the 493 patients with AET < 4%, Lyon Consensus–based stratification yielded 75 (15.2%) reflux hypersensitivity, 95 (19.3%) functional heartburn, 173 (35.1%) results against reflux, and 150 (30.4%) undetermined cases. In particular, 40 of 173 patients (23.1%) in the results against reflux subgroup were labeled as anomalies by the proposed model, suggesting a small but

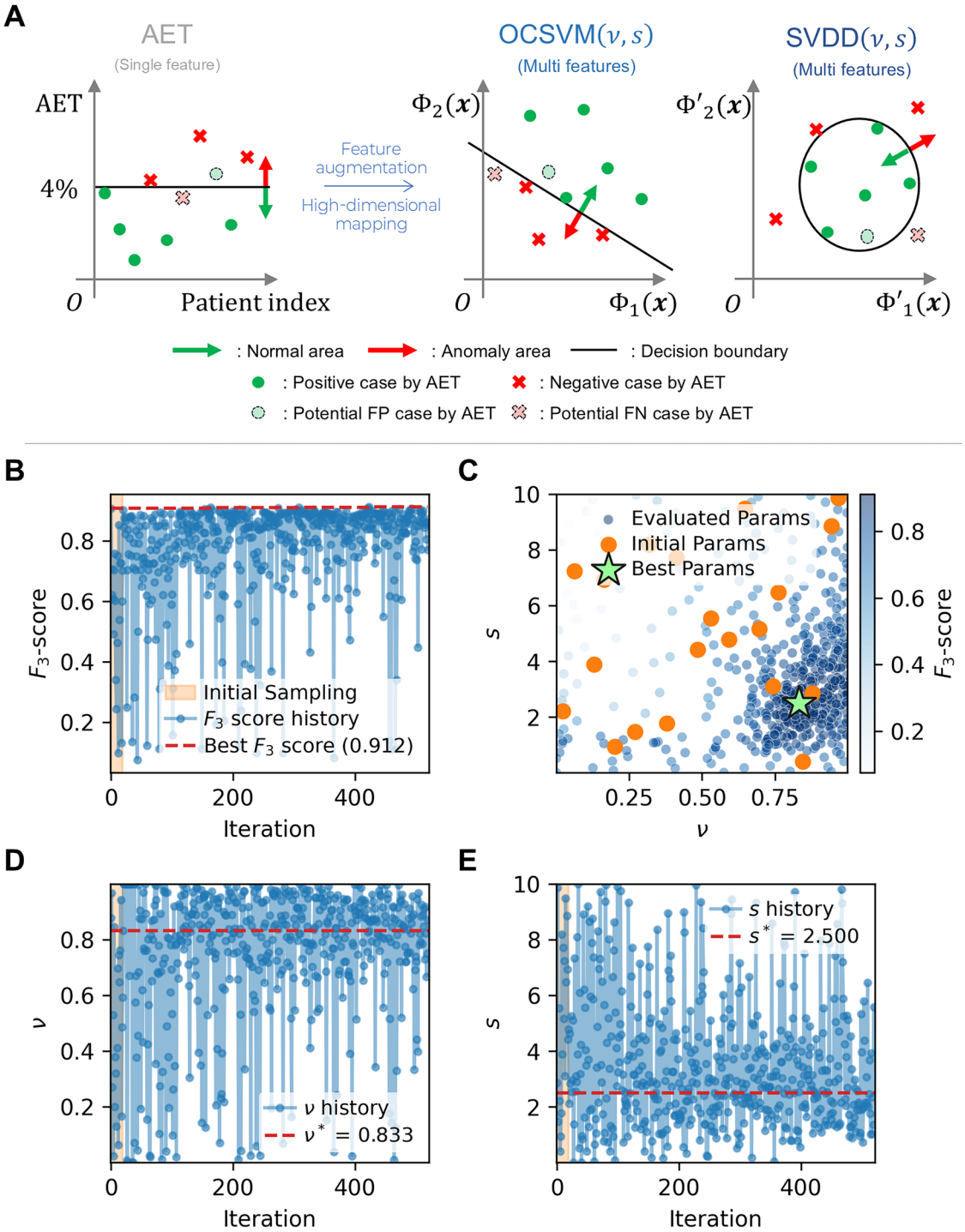


Fig. 4 | Schematic illustration of decision boundary formation and hyperparameter optimization results using BO. A Comparison of decision boundaries for traditional AET-based diagnosis (single feature), OCSVM (multi-feature), and SVDD (multi-feature). **B** Evolution of the F_3 score across BO iterations, with the best score (0.912) highlighted. **C** Visualization of the explored hyperparameter space (ν and s), displaying initial parameters (orange), evaluated parameters (blue), and the optimal set (green star). **D** Convergence history of the hyperparameter ν, with the optimal value determined as $\nu^* = 0.833$. **E** Convergence history of the hyperparameter s, with the optimal value determined as $s^* = 2.500$.

clinically relevant subset of potential false-negative candidates within the AET-negative cohort; detailed stratification and performance metrics are provided in Supplementary Note 3.

Figure 5 summarizes how the proposed AD models identified anomalies compared with the conventional AET-based classification for both the training and test sets, and it illustrates the contribution of each feature to the OCSVM predictions using SHAP analysis. Using the conventional AET-based classification as a baseline, both the OCSVM and SVDD models increased the AD rate by approximately 1.9 times in the training set (Fig. 5A) and the test set (Fig. 5B), respectively. In addition, an independent expert clinician, blinded to the model outputs, retrospectively reviewed each 24-h MII-pH report (including AET, DMS, reflux episode counts, MNBI, SI/SAP, and waveform patterns) to assign GERD suspicious versus non-suspicious labels, which were used as a retrospective reference

for evaluating the intended use case of the study. Relative to this reference, the proposed AD model showed accuracy, precision, recall, and F_1 score of 0.951, 0.867, 1.000, and 0.929 on the test set, compared with 0.870, 1.000, 0.590, and 0.742, respectively, for the conventional AET-based classification (see Supplementary Note 4). These findings suggest that the proposed framework may help identify additional cases for clinical reassessment that are not sufficiently captured by conventional metrics and standard thresholds.

To further analyze the model's decision-making process and clinical relevance, SHAP analysis was applied to the OCSVM model predictions across the full dataset (Fig. 5C). The SHAP beeswarm plot (Fig. 5C, left) illustrates how individual features contribute to AD, showing the distribution and impact of feature values on the model's predictions. Negative SHAP values shift the prediction toward an anomaly, whereas positive values support a normal prediction. Feature values are color-coded from high (red) to low (blue), with deeper red tones indicating higher values of the corresponding feature and deeper blue tones representing lower values. The mean absolute SHAP values (Fig. 5C, right) further quantify each feature's global contribution, highlighting the predominant influence of AET (0.029), followed by kurtosis (0.018) and peak-to-peak amplitude (0.011). In contrast, impulse factor and crest factor demonstrated minimal global contributions.

Table 1 | Confusion matrix comparing AET label and model prediction for total dataset using the OCSVM model

			AET-based label	
		Total	≥ 4%	<4%
	Total patients	612 (100%)	119 (19.4%)	493 (80.6%)
Model prediction	Anomalies	226 (36.9%)	118 (99.2%)	108 (21.9%)
	Non-anomalies	386 (63.1%)	1 (0.8%)	385 (78.1%)

Values are given as *n* (%).

Consistent with clinical definitions, AET showed the most substantial negative SHAP contributions, particularly when feature values were high (red), indicating that overall acid burden remains the dominant feature driving anomaly classification. Similarly, elevated kurtosis also showed strong negative SHAP values, indicating sharper, more peaked signals that often reflect sudden drops in pH below 4. Moreover, these findings provide that sharp waveform deviations provide additional information beyond AET alone. In the case of peak to peak amplitude, both extremely high and low values were linked to negative SHAP contributions, indicating that abnormal classifications arise from either excessive or abnormally muted signal variations. These findings support interpreting the selected features as a recording-level representation that combines conventional acid burden information with higher-order waveform deviations. In this sense, the proposed model does not directly resolve episode-level temporal mechanisms, but it can identify atypical patterns that may not be sufficiently captured by AET alone.

To further quantify the contribution of each feature, an ablation study was performed on the OCSVM model, in which the model was re-optimized via Bayesian hyperparameter search after removing each feature in turn and re-evaluated on the same test set using the expert labels. This analysis confirmed that AET is the single most influential predictor, that kurtosis provides additional discriminative power in borderline weakly acidic cases, and that amplitude-based features (e.g., peak to peak, impulse factor, crest factor) mainly offer redundant or auxiliary information without materially

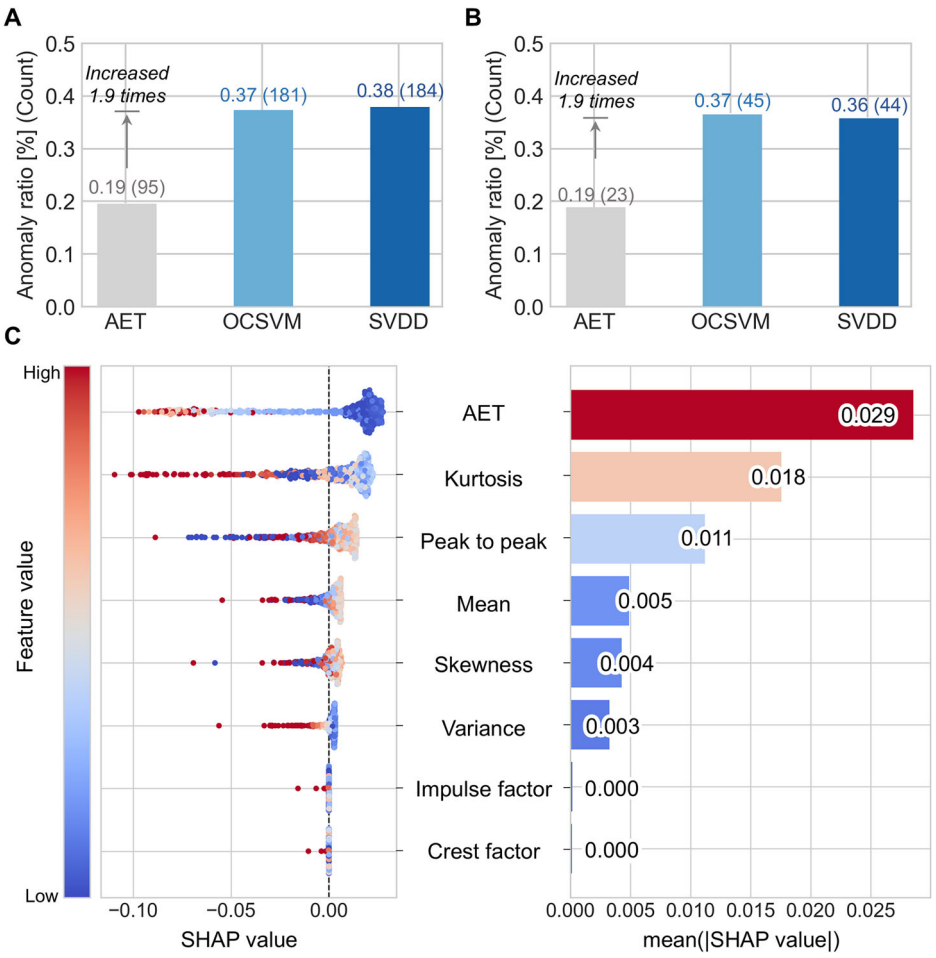


Fig. 5 | Performance comparison of conventional and ML-based AD with SHAP feature interpretation of OCSVM. A Anomaly detection results on the training set, comparing anomaly ratios (bars, %) and counts (numbers in parentheses) detected by conventional AET, OCSVM, and SVDD models. **B** Anomaly detection results on the test set, comparing anomaly ratios (bars, %) and counts (numbers in parentheses) detected by conventional AET, OCSVM, and SVDD models. **C** SHAP summary plot and mean absolute SHAP values illustrating feature contributions to OCSVM predictions.

altering overall performance. Detailed results are provided in Supplementary Note 5.

AD results and interpretation of representative cases

Given that the OCSVM and SVDD models yielded nearly similar results, this section focuses on the outcomes of the OCSVM model for clarity. The results for the SVDD model are provided in Supplementary Note 6. The OCSVM model's behavior in the multi-feature space and the positioning of individual cases relative to the decision boundaries were examined. Figure 6A visualizes the OCSVM AD outcomes by projecting the test dataset into a two-dimensional principal component (PC) space using PCA, based on eight extracted features: mean, variance, skewness, kurtosis, peak to peak, impulse factor, crest factor, and AET. The loading coefficients for the first PC (PC1) were −0.054, 0.034, 0.028, -0.042, 0.015, 0.023, 0.018, and 0.996, while those for the second PC (PC2) were 0.055, -0.024, -0.16, 0.977, 0.114, -0.001, -0.002, and 0.048, respectively.

Each point denotes a patient, with the background color indicating the OCSVM-derived anomaly score (blue for normal and red for anomalous). Most normal cases form a blue-shaded cluster near the center, whereas anomalous cases are more widely dispersed toward the red regions with negative scores. Triangles represent anomalies detected by both AET and OCSVM, while squares denote additional anomalies identified solely by OCSVM. These additional detections, enabled by the incorporation of diverse features and a nonlinear decision boundary, show the model's capacity to identify subtle or complex patterns that are overlooked by the AET-based method.

Four representative patient cases, previously presented in Fig. 2, are highlighted with star markers in Fig. 6A and examined in more detail in Fig. 6B–E. Patient a represents a TP, correctly identified as abnormal by both the AET-based method and the proposed model. Patient b is a TN, consistently classified as normal by both methods. On the contrary, patients c and d are labeled as normal based on the AET threshold but were detected as anomalies by the proposed OCSVM model. Figure 6B–E present three classification perspectives: the traditional method based on AET values, the proposed method using OCSVM anomaly scores, and auxiliary metrics such as DMS or total reflux. In addition, SHAP analysis is provided to enhance interpretability by explaining why the proposed model classified a case as normal or anomalous. The corresponding signal patterns and their clinical relevance are discussed in detail in the following sections.

Representative TP and TN cases are shown in Fig. 6B, C, with detailed signal profiles presented in Fig. 2A, B. In the TP case (Patient a; Fig. 2A), prolonged acid exposure is clearly observed, resulting in a notably high AET (19.98) and a pathological DMS (82.50), both indicative of conclusive GERD. This case was correctly identified as anomalous by the OCSVM model, primarily due to strong negative SHAP contributions from AET and kurtosis, resulting in an anomaly score of −0.097 (Fig. 6B). In contrast, the TN case (Patient b; Fig. 2B) showed minimal acid reflux, with an AET of 0.00 and a DMS of 0.80, both well within normal ranges. The model assigned a positive anomaly score of 0.025, influenced by positive SHAP contributions from AET and skewness, which correctly aligned with a normal classification (Fig. 6C).

Two cases representing potential FNs are depicted in Fig. 6D, E, with detailed signal profiles shown in Fig. 2C, D. Although these cases were originally categorized as normal by conventional AET criteria, the proposed OCSVM model identified them as anomalous. In the first potential FN case (Patient c; Fig. 2C), frequent but short-lived acid reflux episodes were observed, resulting in an AET below the diagnostic threshold (3.39). However, the DMS (16.90) exceeded the pathological cutoff, and the report documented a total of 103 reflux episodes (76 upright and 27 recumbent), suggesting that this case may merit additional clinical review despite being categorized as normal by the conventional AET criterion. The model classified this instance as anomalous (anomaly score: −0.004), influenced primarily by SHAP contributions from peak to peak amplitude and variance (Fig. 6D).

The second potential FN case (Patient d; Fig. 2D) exhibited a pattern characterized by transient pH dips and high signal variability, with low values of AET (0.58) and DMS (2.40). Despite these seemingly normal metrics, the model assigned a negative anomaly score (−0.088), driven by significant SHAP contributions from kurtosis and skewness (Fig. 6E). Clinical records indicated a total of 59 reflux episodes (37 upright and 22 recumbent) and 695 belching events, 284 of which were temporally associated with reflux (SI 40.9%, SAP 0.0%), indicating a substantial symptom burden that is not fully reflected by AET and DMS alone and may support further clinical reassessment.

A detailed analysis of two additional potential FN cases, similarly overlooked by conventional AET or DMS criteria yet identified as anomalous by the proposed model, is presented in Supplementary Note 7. These cases illustrate that some patients categorized as normal by conventional AET or DMS criteria may still exhibit signal patterns and auxiliary clinical features that are not fully reflected by those metrics alone. While extended monitoring or prospective confirmation would be required for definitive disease-level interpretation, these examples suggest that the proposed model may help identify additional cases for clinical reassessment within diagnostically ambiguous settings.

Discussion

This study presents a data-driven clinical reassessment framework for patients with AET < 4%, rather than a direct diagnostic model for GERD itself. The target of the proposed framework is to identify potential false-negative candidates who may be overlooked by the conventional AET criterion, and the task is to detect subtle signal patterns that may support further clinical reassessment. To this end, we combined the conventional clinical index AET with statistical and waveform-derived features extracted from 24-h MII-pH signals and used them in an unsupervised AD framework. In this intended use-case, the proposed model is designed not to replace the conventional AET-based classification, but to provide additional quantitative information that may help clinicians prioritize reassessment in diagnostically ambiguous cases.

Model development was guided by four key considerations. First, the dataset showed a strong class imbalance, with much more normal than abnormal samples, rendering supervised learning approaches less effective. Second, in accordance with the Seoul Consensus Recommendation, all cases with AET ≥ 4% were labeled as abnormal, including both inconclusive and conclusive categories. Third, features such as kurtosis, variance, and peak-to-peak amplitude were incorporated to better capture recording-level waveform deviations associated with transient reflux-like patterns that may not be sufficiently reflected by AET alone. Finally, due to the heterogeneity of signal anomalies and the absence of detailed annotations, an unsupervised approach was chosen, allowing the model to learn the distribution of normal patterns and identify deviations without relying on labeled data.

Our model showed practical utility as a reassessment aid for identifying potential FN candidates, identifying nearly twice as many anomalies as the conventional AET-based criterion and achieving an F_3 score of approximately 0.9. Representative case analyses further showed that the proposed framework can identify borderline or weakly acidic patterns that are not sufficiently reflected by AET or DMS alone and may therefore warrant additional clinical review.

SHAP analysis provided additional interpretive support for the selected feature set at the recording level. High AET values consistently showed strong negative SHAP contributions, reaffirming their consistency with established clinical definitions of abnormal reflux. Additionally, data-driven statistical and waveform features, such as kurtosis and peak-to-peak amplitude, significantly influenced anomaly predictions. This multi-dimensional feature set enabled the effective detection of atypical reflux-like signal patterns that were not sufficiently reflected by AET alone. However, these selected features should be interpreted not as direct markers of reflux-specific pathophysiological mechanisms, but rather as a compact representation of acid burden and waveform variability that may help identify atypical cases for further clinical reassessment.

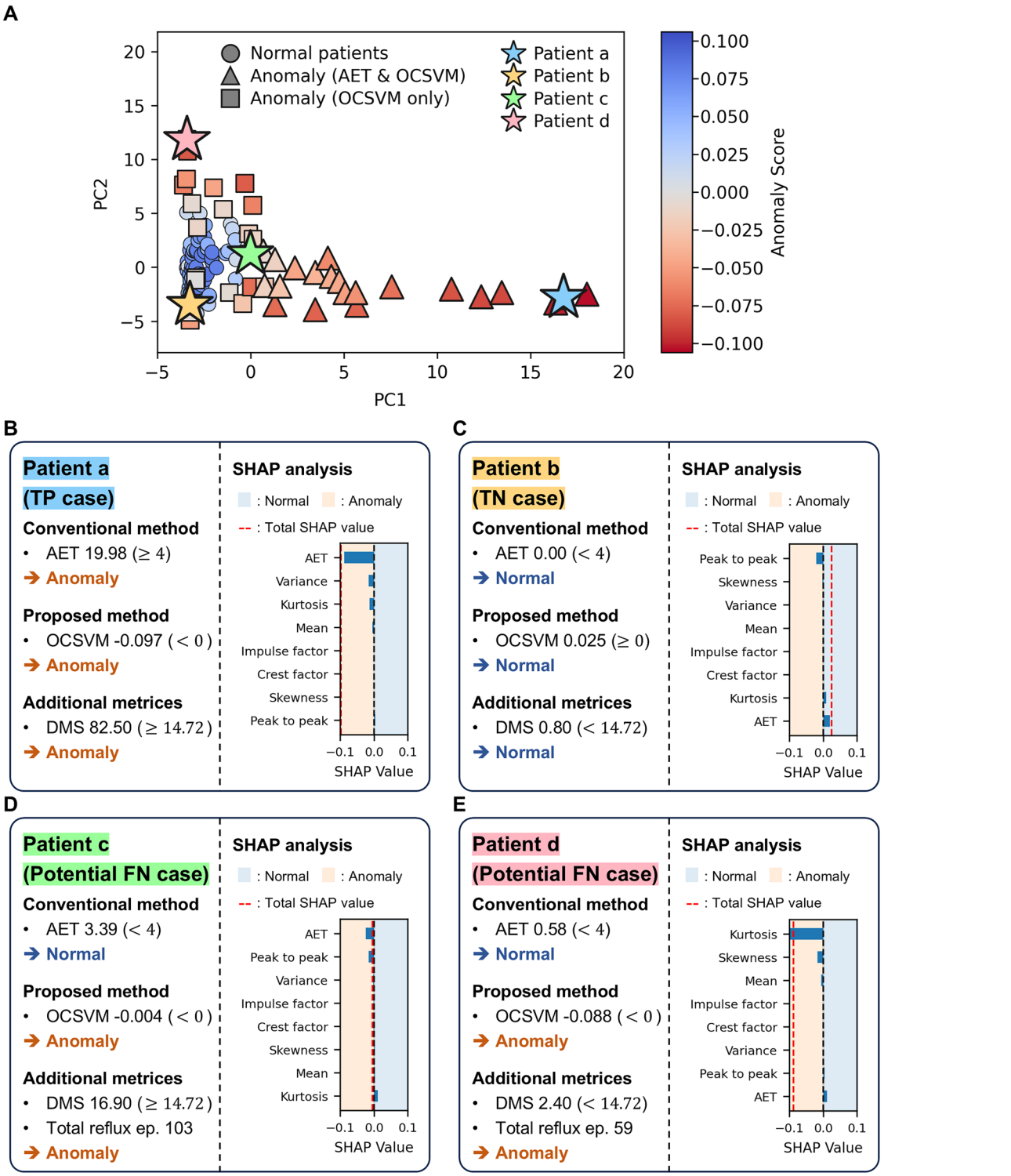


Fig. 6 | Comparison of AD outcomes and SHAP-based interpretation of representative patient cases. A Principal component visualization comparing OCSVM AD with AET classification, with stars marking four representative patients (a–d). **B** Patient a (TP): Both AET and OCSVM detect an anomaly, **C** Patient b (TN): Both AET and OCSVM detect normal. **D** Patient c (Potential FN): Missed by AET but detected as an anomaly by OCSVM. **E** Patient d (Potential FN): Missed by AET but detected as an anomaly by OCSVM.

The proposed AD framework was evaluated under label noise, sensor artifacts, and inter-patient variability, without requiring pre-labeled anomaly subtypes. A comparative study with various unsupervised AD models (see Supplementary Note 8) indicated that the proposed multi-feature kernel-based models, OCSVM and SVDD, performed favorably for identifying potential false negatives within the specific context of our dataset and problem formulation. While deep learning-based approaches showed limited performance in this specific setting, likely due to data scale and complexity constraints, they may become more useful for modeling temporal dependencies in future studies. Specifically, if larger datasets or

advanced feature extraction techniques (e.g., STFT, wavelet transform) become available, such models may provide a more useful basis for exploring temporal dependencies.

The present findings should be interpreted within the intended scope of the proposed framework. In this study, the anomaly score is best understood as a recording-level signal that may highlight additional cases for clinical reassessment beyond the conventional AET criterion, rather than as a direct indicator of reflux-specific pathophysiological mechanisms. In addition, the independent expert review provided a practical retrospective reference for evaluating the intended use-case of the model under real-world constraints, although further validation with repeated monitoring and prospective follow-up would strengthen the clinical interpretation of these findings. Because the current study was based on a retrospective single-center cohort with 24-h MII-pH signals, it was not designed to directly resolve episode-level temporal mechanisms or to distinguish all possible sources of anomaly signals. In addition, because routine esophageal biopsy is not standard practice in the workup of GERD in Korea, eosinophilic esophagitis could not be completely excluded in a subset of patients, although its relatively low prevalence in Korean populations makes a substantial impact on the present results less likely. A more detailed discussion of these limitations and future directions is provided in Supplementary Note 9.

In future work, this framework could be validated on longer (48–72 h) wireless MII-pH signals and multi-center datasets, enabling more detailed analyses of the relationships between esophageal acid exposure, symptom frequency, and the anomaly diagnosis by the proposed model, using approaches similar to those adopted in related works[17,18]. Further, deep learning-based models could be applied to raw multichannel MII signals combined with time–frequency preprocessing (e.g., STFT, wavelet), as in related works[14,15] to complement the present statistical features, better exploit their spatiotemporal information, and improve clinical decision support. Such future studies may enable more direct modeling of episode-level temporal dynamics, including clustering, day–night variation, and other reflux-relevant temporal patterns that are not explicitly resolved in the present framework.

Methods

Medical records and ethical considerations

The data of MII-pH monitoring for this study were retrospectively obtained from Severance Hospital, Seoul, Korea, which is a tertiary referral hospital in Korea. The study population consisted of patients referred from primary or secondary care facilities to a tertiary referral center due to refractory or suboptimal response to standard GERD treatment, who subsequently underwent 24-h MII-pH monitoring between October 2018 and July 2022. A total of 716 consecutive patients were initially screened. Of these, 628 patients met the following inclusion criteria: (1) normal upper gastrointestinal endoscopy performed within 6–12 months prior to or after MII-pH monitoring, confirming the absence of structural abnormalities including esophageal or gastric malignancy; (2) normal esophageal manometry, which was performed routinely prior to MII-pH monitoring to exclude primary esophageal motility disorders including achalasia, and to precisely localize the lower esophageal sphincter for accurate catheter placement; and (3) MII-pH monitoring performed in the PPI-off state. The remaining 88 patients were excluded due to one or more of the following: abnormal manometry findings (including achalasia and other motility disorders), endoscopic abnormalities, or ongoing PPI use at the time of monitoring. After further exclusion of 16 patients with insufficient signal quality or incomplete recording data, the final analytic cohort consisted of 612 patients.

This study was conducted in accordance with the Declaration of Helsinki. The study protocol was approved by the Institutional Review Board (IRB) of Gangnam Severance Hospital, Yonsei University College of Medicine (IRB No. 3-2022-0221). Given the retrospective nature of the study, the requirement for written informed consent was waived by the IRB.

Dataset description

The wired MII-pH monitoring (ZepHr Impedence/pH, Diversatek Healthcare) dataset was collected from patients at Severance Hospital and includes six impedance signals and one pH signal. The impedance signals are used to determine the direction of liquid movement along the esophagus, while the pH signal measures acidity levels, allowing for the detection of both acid and nonacid reflux episodes. AET and DMS values, obtained from the *Zvu® Advanced GI Diagnostic Software* (Version 3.0.0, Diversatek Healthcare, Milwaukee, WI, USA), served as reference diagnostic labels in this study. These values were automatically calculated by the commercial software based on the acquired raw signals. Because the internal procedures for noise handling and calculation are not fully disclosed, AET was recalculated from the raw pH signals and these recalculated values were used in our analyses.

Patients were categorized based on AET readings into three groups: Normal (0–4%, label 0), Inconclusive (4–6%, label 1), and Conclusive (≥6%, label 2). Concurrently, the DMS provides further classification into Normal, Mild, Moderate, and Severe categories, with a threshold of 14.72 indicating abnormal conditions[19]. Clinical information of MII-pH monitoring dataset is provided in Table 2. A total of 612 patients, 250 males, 344 females, and 18 of unknown gender, were included in the study, revealing a significant class imbalance, with 80.6% (493/612) classified as Normal and 19.4% (119/612) as either Inconclusive or Conclusive. The total dataset ($n = 612$) was divided into a training set ($n = 489$; 80% of the total dataset) and a test set ($n = 123$; 20%). In alignment with Recommendation 13 of the Seoul Consensus[1], cases with AET values ≥ 4%, including both the inconclusive and conclusive categories, were explicitly labeled as anomalies.

Feature extraction and selection

To characterize the esophageal acid exposure signal, we extracted 10 additional data-driven statistical and waveform features, along with the traditional AET, from the 24-h pH signals in the datasets. AET is computed using data from six impedance sensors and one pH sensor for each patient. This measurement reflects the percentage of time during which esophageal pH remains below 4, as shown in Eq. 1 To enhance specificity for GERD-related anomalies, AET calculation is performed only when the difference between the last impedance signal, I_6, and the other impedance signals is greater than or equal to a factor of $c = 500\ \Omega$, indicating significant reflux.

$$AET = \left(\frac{N}{N_T}\right) \times 100(\%) \tag{1}$$

where N_T denotes the total number of data points, N represents the number of data points with $pH \le 4$, and the condition $\left(I_m - I_6\right) \ge c$ for any $m = 1, 2, 3, 4, 5$ is satisfied, where $c = 500\ \Omega$.

Furthermore, additional statistical and waveform-derived features were extracted and combined to identify complex reflux patterns, which include statistical metrics such as mean, variance, and RMS, as well as waveform characteristics such as skewness, kurtosis, and peak to peak amplitude, as presented in Table 3. This enhanced feature extraction aims to improve AD, particularly for cases involving sharp pH fluctuations or atypical reflux patterns[20–22]. In total, 11 features, including AET, were applied in the analysis.

Model-based AD

In this study, we implemented model-based AD techniques using OCSVM[23] and SVDD[24] to classify recording-level feature patterns extracted from 24-h MII-pH signals as normal-like or anomalous for reassessment purposes. The SVM, from which these models are derived, is fundamentally a ML technique that identifies an optimal hyperplane capable of categorizing different data classes within a feature space.

OCSVM is an adaptation of the original SVM specifically designed for AD. It distinguishes between normal and anomalous data points by separating them from the origin in a transformed high-dimensional feature space. Unlike traditional SVM, which is a supervised method, OCSVM

Table 2 | Clinical information of MII-pH monitoring dataset

	AET			
	<4%	4–6%	≥6%	Total
Total	493 (80.6%)	36 (5.9%)	83 (13.6%)	612 (100%)
Male	184	20	46	250
Female	293	16	35	344
Unknown	16	0	2	18
Age, means.d.	46.3 ± 22.4	39.8 ± 27.8	31.9 ± 27.7	44.0 ± 24.0
Training set	393 (80.4%)	30 (6.1%)	66 (13.5%)	489 (79.9%)
Test set	100 (81.3%)	6 (4.9%)	17 (13.8%)	123 (20.1%)

Values are given as *n* (%).

Table 3 | Features and formulas used for feature extraction

Features	Formula	Features	Formula
Absolute mean	$\frac{1}{n}\sum_{k=1}^{n}\lvert x_k\rvert$	Peak to peak	$\max(X)-\min(X)$
Mean	$\frac{1}{n}\sum_{k=1}^{n}x_k$	RMS	$\sqrt{\frac{1}{n}\sum_{k=1}^{n}x_k^2}$
Variance	$\frac{1}{n}\sum_{k=1}^{n}(x_k-\mu_X)^2$	Impulse factor	$\frac{\max(\lvert X\rvert)}{\text{Absolute mean}}$
Skewness	$\frac{1}{n}\sum_{k=1}^{n}\left(\frac{x_k-\mu_X}{\sigma_X}\right)^3$	Shape factor	$\frac{\text{RMS}}{\sqrt{\text{Absolute mean}}}$
Kurtosis	$\frac{1}{n}\sum_{k=1}^{n}\left(\frac{x_k-\mu_X}{\sigma_X}\right)^4-3$	Crest factor	$\frac{\max(\lvert X\rvert)}{\text{RMS}^2}$

operates under unsupervised conditions and does not require labeled data. This characteristic is particularly beneficial in medical applications, where labeled data may be scarce or incomplete. The mathematical model of OCSVM focuses on maximizing the separation margin, ρ, and is formulated as the following optimization problem:

$$\begin{aligned}&\min_{\boldsymbol{w}\in\mathbb{R}^q,\xi_i\in\mathbb{R},\rho\in\mathbb{R}}\ \tfrac{1}{2}||\boldsymbol{w}||_2^2+\tfrac{1}{vn}\sum_{i=1}^{n}\xi_i-\rho\\ &subject\ to\big(\boldsymbol{w}\cdot\boldsymbol{\Phi}(\boldsymbol{x}_i;s)\big)\geq\rho-\xi_i,\\ &\xi_i\geq 0,\\ &i=1,\ldots,n,\end{aligned}\tag{2}$$

where $\boldsymbol{w}$ denotes the normal vector to the hyperplane, n represents the number of training data points, and ξ_i indicate slack variables that allow some data points to lie within the margin. $-\rho$ is an offset that optimizes the position of the hyperplane. $\Phi(\boldsymbol{x}_i;s)$ represents the mapping function that transforms data into a higher-dimensional space, facilitating the linear separation of nonlinear data. The mapping $\boldsymbol{\Phi}(\boldsymbol{x}_i;s)$ is often replaced by a kernel function, which allows the results of higher-dimensional computations to be obtained without directly calculating them. Specifically, the Gaussian kernel is widely used for nonlinear problems, with s representing the variance of the Gaussian kernel.

The hyperparameters v and s play a crucial role in shaping the decision boundaries of the OCSVM. The hyperparameter v, which ranges from 0 to 1, controls the proportion of support vectors. A larger v allows for more anomalies. The parameter s influences the width of the kernel function. Smaller values of s, resulting in a narrow kernel, enable the model to create more complex and fine-grained decision boundaries, while larger values of s, corresponding to a wider kernel, lead to smoother and less complex decision boundaries.

SVDD offers an alternative approach by encapsulating data within a hypersphere instead of separating it with a hyperplane. Its goal is to define the smallest possible sphere that encompasses all normal data points, thereby facilitating the detection of outliers or anomalies. The SVDD model achieves this by minimizing the radius R of the hypersphere, formulated as the following optimization problem:

$$\begin{aligned}&\min_{\boldsymbol{a}\in\mathbb{R}^q,\xi_i\in\mathbb{R},R\in\mathbb{R}}\ R^2+\frac{1}{vn}\sum_{i=1}^{n}\xi_i subject\ to\ ||\boldsymbol{\Phi}\prime(\boldsymbol{x}_i;s)-\boldsymbol{a}||_2^2\leq R^2\\ &+\xi_i,\ \xi_i\geq 0,\ i=1,\ldots,n,\end{aligned}\tag{3}$$

where R represents radius of the hypersphere including the normal data, and $\boldsymbol{a}$ indicates center of the hypersphere in the multidimensional feature space. The mapping function $\boldsymbol{\Phi\prime}(\boldsymbol{x}_i;s)$, along with the parameters n and ξ_i is identical to those used in the OCSVM formulation. Similarly, the hyperparameter v and s serve the same role as in OCSVM, controlling the trade-off between model sensitivity and generalization. By appropriately adjusting these hyperparameters, both models can effectively shape decision boundaries or enclosures within the multidimensional feature space, thereby allowing for more nuanced and accurate identification of abnormal reflux patterns in patients.

Evaluation metrics

The performance of AD models in this study was evaluated using the F_β score, with the GS labels defined by the AET 4% criterion. TPs and TNs denote correctly identified anomaly and normal cases, respectively, whereas FNs and false positives (FPs) correspond to incorrectly identified anomaly and normal cases. Precision (P) is defined as the ratio of TPs to the sum of TPs and FPs, and recall (R) is defined as the ratio of TPs to the sum of TPs and FNs.

The F_β score integrates these measures into a single metric as defined in Eq. 4. This metric provides an adjustable balance between P and R through the β parameter, where R is considered β times as important as precision. In this study, as accurately predicting positive cases, i.e., anomalies, is essential with the new method, greater emphasis should be placed on R. Therefore, the F_3 score, which corresponds to the F_β score with $\beta=3$, is employed to assign higher weight to recall.

$$F_\beta=\frac{1}{\frac{1}{1+\beta^2}\cdot\frac{1}{P}+\frac{\beta^2}{1+\beta^2}\cdot\frac{1}{R}}\tag{4}$$

where $Precision(P)=\frac{TP}{TP+FP}$, *and* $Recall\,(R)=\frac{TP}{TP+FN}$.

Hyperparameter tuning

Both the OCSVM and SVDD models are inherently unsupervised and require careful tuning of their hyperparameters, particularly v and s, to optimize their AD performance. Since the available training dataset contains explicit labels differentiating normal from abnormal instances, these labels can effectively guide the selection of hyperparameters that facilitate accurate classification of anomalies.

To efficiently optimize these hyperparameters, BO is utilized—a technique renowned for its effectiveness in navigating complex parameter spaces[25,26]. BO systematically models the relationship between the hyperparameters and the corresponding model performance, allowing for the identification of an optimal hyperparameter set, $\boldsymbol{\theta}^*$, that maximizes the selected evaluation metric, specifically the F_3 score. This optimization process is formulated as Eq.: 5

$$Objective:\ arg\max_{\boldsymbol{\theta}}F_3(AET\ Class,\ g(\boldsymbol{X}|\boldsymbol{\theta}))\tag{5}$$

where $g(\boldsymbol{X}|\boldsymbol{\theta})$ represents the predicted class label generated by the AD model (either OCSVM or SVDD), $\boldsymbol{\theta}=(v,s)$ denotes the set of hyperparameters to be optimized, and $\boldsymbol{\theta}^*$ indicates the optimized set. The hyperparameter v controls the fraction of outliers, and s corresponds to the kernel bandwidth in the RBF kernel.

In our implementation, the surrogate model within BO is a Gaussian process regression employing a Matérn 5/2 kernel. The EI acquisition

function is used to determine subsequent hyperparameter candidates, v and s, by balancing exploration of new regions in the hyperparameter space with exploitation of known high-performing regions, which means the F_3 score here. Initially, the BO procedure is seeded with 20 data points generated using LHS, ensuring a thorough, space-filling exploration of the defined hyperparameter bounds. During each BO iteration, the candidate pair (v and s) is evaluated via 5-fold cross-validation on the training set, each fold yielding an F_3 score. The mean F_3 score across all five folds is used as the objective value. This 5-fold cross-validation ensures that v and s are tuned not only to fit the training data but also to generalize across unseen folds, thereby preventing overfitting. This optimization process is critical for developing a robust model capable of identifying not only the known anomalies in the training dataset but also potential unseen anomalies, thereby enhancing the generalizability and clinical utility of the AD framework.

Feature importance using SHAP

In clinical applications, the reliability of ML models relies not only on their predictive accuracy but also on their transparency and interpretability. This critical need for clear insight in medical decision-making is met through explainable artificial intelligence (XAI). Among various XAI techniques, SHAP values are commonly employed to illustrate how different input variables impact a model's predictions[27].

SHAP values quantify each feature's contribution to a specific prediction relative to the average prediction made across all possible combinations of features. The SHAP value, ϕ_i, for a particular feature is mathematically defined as follows:

$$\phi_i = \sum_{S \subseteq F/\{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \left(f(S \cup \{i\}) - f(S)\right) \quad (6)$$

where i denotes the feature of interest, S indicates a subset of all features excluding the feature of interest, and F represents the set of all features. $f(S)$ stands for the model's prediction when only the features in subset S are considered, while $f(S \cup \{i\})$ denotes the model's prediction when feature i is added to subset S.

This approach not only highlights the most influential features but also illustrates how the presence or absence of a feature affects the prediction, providing an in-depth understanding of feature interactions and dependencies. By applying SHAP values, the model's decision-making process can be analyzed, offering insights into why certain cases are identified as anomalies. Such an explanation is essential for clinical acceptance and trust, as it enables healthcare professionals to understand and validate the model's decisions, ensuring alignment with clinical standards and requirements[28].

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to the sensitive nature of the clinical data, which includes identifiable participant information, and because public dissemination would breach patient privacy and violate the terms of the IRB approval under which the data were collected, but are available from the corresponding author on reasonable request.

Code availability

The code developed in this study is available from the corresponding author upon reasonable request.

Received: 16 September 2025; Accepted: 15 May 2026;
Published online: 27 May 2026

References

1. Jung, H. K. et al. 2020 Seoul consensus on the diagnosis and management of gastroesophageal reflux disease. *J. Neurogastroenterol. Motil.* **27**, 453–481 (2021).
2. Batista, A., de, O. & Dantas, R. O. Correlation between symptoms and reflux in patients with gastroesophageal reflux disease. *Arq. Gastroenterol.* **59**, 184–187 (2022).
3. Richter, J. E. & Rubenstein, J. H. Presentation and epidemiology of gastroesophageal reflux disease. *Gastroenterology* **154**, 267–276 (2018).
4. Argüero, J. & Sifrim, D. Pathophysiology of gastro-oesophageal reflux disease: implications for diagnosis and management. *Nat. Rev. Gastroenterol. Hepatol.* **21**, 282–293 (2024).
5. Delaney, B. C. Review article: prevalence and epidemiology of gastro-oesophageal reflux disease. In *Alimentary Pharmacology and Therapeutics, Supplement* vol. 20, 2–4 (Blackwell Publishing Ltd, 2004).
6. Yamasaki, T., Hemond, C., Eisa, M., Ganocy, S. & Fass, R. The changing epidemiology of gastroesophageal reflux disease: are patients getting younger? *J. Neurogastroenterol. Motil.* **24**, 559–569 (2018).
7. Hussain, Z. H. et al. The proton pump inhibitor non-responder: a clinical conundrum. *Clin. Transl. Gastroenterol.* **6**, p e106 (2015).
8. Jiang, Y., Jiang, L., Ye, B. & Lin, L. Value of adjunctive evidence from MII-pH monitoring and high-resolution manometry in inconclusive GERD patients with AET 4–6%. *Ther. Adv. Gastroenterol.* **14**, 17562848211013484 (2021).
9. Neto, R. M. L., Herbella, F. A. M., Schlottmann, F. & Patti, M. G. Does DeMeester score still define GERD? *Dis. Esophagus* **32**, doy118 (2019).
10. Prakash Gyawali, C. et al. Modern diagnosis of GERD: the Lyon consensus. *Gut* **67**, 1351–1362 (2018).
11. Fernando, T., Gammulle, H., Denman, S., Sridharan, S. & Fookes, C. Deep learning for medical anomaly detection a survey. *ACM Comput. Surv.* **54**, 141 (2022).
12. Dairi, A., Harrou, F. & Sun, Y. Deep generative learning-based 1-SVM detectors for unsupervised COVID-19 infection detection using blood tests. *IEEE Trans. Instrum. Meas.* **71**, 2500211 (2022).
13. Hojjati, H. & Armanfard, N. DASVDD: deep autoencoding support vector data descriptor for anomaly detection. *IEEE Trans. Knowl. Data Eng.* **36**, 3739–3750 (2024).
14. Lomoio, U. et al. A convolutional autoencoder framework for ECG signal analysis. *Heliyon* **11**, e41517 (2025).
15. Hu, D. et al. FECT-OSA: a transformer-enhanced multimodal system for non-contact sleep apnea monitoring. *Alex. Eng. J.* **128**, 628–641 (2025).
16. Gong, H., Li, Y., Zhang, J., Zhang, B. & Wang, X. A new filter feature selection algorithm for classification task by ensembling pearson correlation coefficient and mutual information. *Eng. Appl. Artif. Intell.* **131**, 107865 (2024).
17. Gardner, J. D. The relationship between esophageal acidity and symptom frequency in symptomatic nonerosive gastroesophageal reflux disease. *Physiol. Rep.* **10**, e15442 (2022).
18. Gardner, J. Oscillating esophageal acid sensitivity in symptomatic reflux hypersensitivity and functional heartburn. Preprint at https://doi.org/10.32388/IJUE1J.2 (2022).
19. Liu, S. et al. Research on gastroesophageal reflux disease based on dynamic features of ambulatory 24-hour esophageal pH monitoring. *Comput. Math. Methods Med.* **2017**, 9239074 (2017).
20. Ghisa, M. et al. The lyon consensus: does it differ from the previous ones? *J. Neurogastroenterol. Motil.* **26**, 311–321 (2020).
21. Kim, S., Noh, Y., Kang, Y., Park, S. & Ahn, B. Fault classification model based on time domain feature extraction of vibration data. *J. Comput. Struct. Eng. Instit. Korea* **34**, 25–33 (2021).
22. Mochammad, S., Kang, Y. J., Noh, Y., Park, S. & Ahn, B. Stable hybrid feature selection method for compressor fault diagnosis. *IEEE Access* **9**, 97415–97429 (2021).

23. Hui, K. H., Ooi, C. S., Lim, M. H., Leong, M. S. & Al-Obaidi, S. M. An improved wrapper-based feature selection method for machinery fault diagnosis. *PLoS ONE* **12**, e0189143 (2017).
24. Schölkopf, B. et al. *Estimating the Support of a High-Dimensional Distribution*. http://direct.mit.edu/neco/article-pdf/13/7/1443/814849/089976601750264965.pdf.
25. Tax, D. M. J. & Duin, R. P. W. Support vector data description. *Mach. Learn.* **54**, 45–66 (2004).
26. Lee, J. et al. Machine learning-based inverse design methods considering data characteristics and design space size in materials design and manufacturing: a review. *Mater. Horiz.* **10**, 5436–5456 (2023).
27. Park, K. et al. Designing staggered platelet composite structure with Gaussian process regression based Bayesian optimization. *Compos. Sci. Technol.* **220**, 109254 (2022).
28. Lundberg, S. M., Allen, P. G. & Lee, S.-I. *A Unified Approach to Interpreting Model Predictions*. https://github.com/slundberg/shap.

Acknowledgements

This work was supported by the 2025 Nanomedical Devices Development Project of the National Nanofab Center (NNFC). Additional support was provided by the Development of a Strategic Platform to Support Bio-Semiconductor Convergence Technologies and Services Program of NNFC and by the InnoCORE program funded by the Ministry of Science and ICT (MSIT) (N10250154). This research was also supported by the National Research Foundation of Korea (NRF) grants funded by the Korean government (MSIT) (RS-2024-00439931 and RS-2025-16070951), and by the National Research Council of Science & Technology (NST) grant funded by the Korean government (MSIT) (No. GTL25061-000). This study was further supported by a grant from the Korean Society of Neurogastroenterology and Motility (KSNM-24-01; Hee Man Kim). The funders had no role in the study design; data collection, analysis, or interpretation; manuscript preparation; or the decision to publish.

Author contributions

S.L. conceptualized the study, developed the methodology, conducted the investigation and software implementation, drafted the original manuscript, and contributed to its revision. J.L. contributed to the development of the methodology and software and co-wrote and revised the manuscript. D.P. supervised the methodology and co-wrote and revised the manuscript. S.K.L. curated the data, participated in the investigation and software tasks, and contributed to manuscript revision. J.H.C. contributed to data curation and validation. K.G.L. provided scientific input and assisted in manuscript preparation and revision. H.M.K. contributed to data curation and clinical investigation, drafted the clinical sections of the manuscript, and supported revisions. S.R. supervised the project, performed formal analysis and validation, oversaw project administration, and revised the manuscript. All authors reviewed and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41746-026-02796-y.

Correspondence and requests for materials should be addressed to Kyoung G. Lee, Hee Man Kim or Seunghwa Ryu.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026